

ActInf Textbook Group ~ Cohort 4 ~ Meeting 4 (Chapter 1 part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 4 (Chapter 1 part 2) | 56:07

all right

it's our

second discussion in cohort 4 of chapter

one

uh does anyone at the beginning want to

just bring up any comment or any

specific question about chapter one

or otherwise we have we can

look at one uh recent stream transcript

okay so Terry um you asked a question

about the generative model the

generative process so maybe just

summarize that question

and then and then we'll see how it

played out recently

uh my question was when we were trying

to create a model we create the

generative model we sort of Define it in

four

over four sort of domains and and and

then at then you define the generative

process

but my if if we you know think of the

Markov blanket as separating B

internal state from the underable

external State well if we're creating a

model

and that model depends on us creating

something that is

in fact on mobile or is there not a

corruption within that

thought process so it creates

an impression that within the context of

if I take my Markov blanket is bad my

skin

then um I uh

I cannot know what is the generative process so there's got to be some flaw in our system of modeling if it depends on creating something we can't do all right awesome and uh when you when you asked that it made me think of the recent

um

stream with Maxwell ramstead now operation and Ali and others

and uh

this highlighted a uh inconsistency in the generative model generative process uh topic Ollie do you want to summarize um

what you what you asked and and what they wrote or said

yeah of course uh well uh we were talking about uh the I mean discrepancy between Maxwell's presentation and also the recent paper because uh in figure one in inner screen paper uh they've indicated generative model as encompassing the whole shebang I mean encompassing the internal States the external States and of course the blanket States so

um nowhere generative process was visible in this diagram so we were discussing with Daniel Sanjeev and others where exactly uh General to process can be manifested uh in this diagram so I basically asked this this question where is the generative process in this diagram and uh surprisingly Maxwell uh somehow disowned the earlier literature and

generative process and basically uh I mean uh he he he just I mean interpreted the generative model in a kind of new more sophisticated way that wouldn't require any separate entity as generative process because General to model basically includes the joint probability of every possible of uh every everything that markup blankets um uh I mean partitions so it it is a joint probability of internal States blanket States and the external States uh but on the other hand when we want to uh model a situation and not just I mean describe a physical phenomena or a physical system then of course we can Define our own generative model as something or let's put it this way as kind of dynamics that we expect the system or the situation the situation we have to follow those Dynamics and so for instance if we want to model a very simple linear behavior of a I don't know mechanical system then of course the Newtonian mechanics or you know the terms the lagrangian mechanics of the system can constitute the generative process of the system so uh yeah that's uh basically the uh gist of Maxwell's argument uh at least as far as I understand it quite a discussion we can kind of parse it out better but the transcripts out there like so um as with everything in the textbook

it's sort of like there's a there's some
internal ontology
and then
um it's somehow aligned with the broader
active Gestalt and for that
you will still it helps to understand
the generating process and the particles
of the agents and so um like to to the
to to the earlier question how can you
model the unknown because you're
modeling something into existence in the
act of modeling
so you can have the real temperature of
the room do statistics on an unobserved
that maps to the observables so in
practice there's no issue but then
people have a lot of
um degrees of interpretation with what
that means and that's the whole and what
that means for different situations
but all that aside in terms of
these two ways to talk about generative
model generative process

[Music]

um
as it's being shown in the inner screen
paper the generative models just the
entire process giving rise to 100 of the
system's level description
including states that generate other
observables
whereas in the textbook they go to some
length and distinction to highlight
the generative model
which has an aboutness of a generative
process
the underlying particular partition
like separating it this way is not

changed

so it's the same topology of the of the

underlying

graph that gives rise to all the

you know formalisms

but it's just kind of it's a truly a

nomenclature question

about where you say okay now these it's

like what yeah

it's just nomenclature question about

describing this

but it's one that always comes up

so

what does anyone think about this it's

kind of a funny random but cool ongoing

topic

uh sorry if I may add uh some additional

comments to what I've just previously

said I mean to my mind this recent

interpretation actually uh makes more

sense because uh one of the

misconceptions around fep uh has always

been how I mean how can we sure about

how to model generative process if the

causes of the observables are as the

name suggests latent States or hidden

States right so if we don't know uh the

exact causes of those observations how

is it possible to model it in a precise

probabilistic

probabilistic terms but in this new way

of uh looking at the generative model

versus generative process well we can

basically I mean somehow

put this issue under the Rock by by

somehow reinterpreting the generative

model as uh including both the external

States and the internal States and only

account and only following the
variational density of the internal
States as

I mean tracing or tracking or not
tracking the external States so that's
basically uh I mean the whole thing the
whole model and and its whole relation
with the quote-unquote reality is
something uh more akin to probably an
instrumentalist
view of reality rather than

I mean the realest uh the uh perspective
because uh here we just used this
theoretical framework only to account
for the parts of the reality or the
behavior with the reality that we can
model

it it looks like Alexa tautology but
that's basically what I understood by
this

a new interpretation uh so that's I am
probably one of the ways to evade those
uh criticisms or misconceptions around
the generative process

thank you Ali very interesting

just to kind of connect this to a
previous discussion in live stream six
one of the early paper-based discussions
this was like described as the tale of
two densities and we probably even use
this meme

still

um but the tale of two densities was the
generative direction from the
temperature in the room hidden State
emitting thermometer readings was the
generative model

a generative Direction it's referring to

the same graph still and the recognition density was like from the sensor data the incoming and so that led to a lot of um well like okay the generative is action the recognition sensory and uh I think what Ali is pointing to is that um if you just kind of say uh well even if it has a million nodes whether one's internal external blanket is relative to whichever one you're talking about so those are not absolute States so you might as well just call the whole thing a generative model still with certain uh components whose sparseness gives them certain properties but there's no need to necessarily distinguish between these two um but still it makes sense to think take it think about an agent-based modeling perspective and about the process that generates so so it's not that it invalidates what the textbook says just from a first thought it's it just sort of expands the scope of what people will probably mean with generative model which is just to say the whole generating architecture of the program probably also closer with generative AI so it's it's um probably a good direction to go but it will still but but this could be thought of as just like agent and Niche well can I yeah yeah this could be a

problem about for me about using
language right so whenever you say there
is a temperature in the hidden external
state
when you say temperature that's a
perception uh you know we can think of
it as a representative of the amount of
energy in that external State and we
perceive it as a core layer of heat uh
so so but but actually we we kind of no
idea what that what that what that is we
use this term temperature uh oh but we
don't know what that means inter if you
think of that you know like the the
physics of information Pro or physics is
information process and so that the
temperature is information that exists
in the all knowable external State we
our perception of temperature is
actually part of the electromagnetic
spectrum you know the
um infrared light is what were is what
we detect as temperature and then we
call we we call it a thing in the x or
we may wait for we we make an assumption
about what that means in the external
state but really I I kind of think that
we uh our our quality of temperature
isn't what we mean when we measure the
temperature
of the external state with a thermometer
or a temperature receptor in our skin
you know so I again I always feel like
I'm talking like I've taken too many
drugs and I'm a 14 year old but you you
know it it
cut this this kind of
um

it feels like there are loads of like
syntaxers in the thought processes
around all this so I'm I'm and
um it it it confuses me

Lexi yeah I was had a perspective on
this question please bear with me if it
sounds a little too simplistic and you
know think about it this way you know
I'm quoting Mark songs by the way if my
body is immersed in an environment with
you know 70 degrees uh Celsius
temperature and I don't solve that
problem I die

if my body is immersed in the
environment with zero degrees Celsius
and I don't solve that problem I die so
at some point it is not about complex
philosophical things because you can go
into physics originally temperature and
entropy were variables designed for a
system at equilibrium

so we can get as semantically complex as
you like but again taken back to bodies
you know there is that very specific
thing that if you you know are outside
of the homeostatic range of temperatures
and you do not solve that problem
originally through reflexes through
sweating or whatever and maybe through
Behavior if you don't solve it you die
so how is that for the you know
temperature is that okay

yeah but just one point on temperature I
think you you

um it's a great question and you you are
indicating like sort of the real
ontological base existence reality which
of course even people have different

views on that but you can model the unobserved state as like we could say this is the the um the SAT score but we're not fully observing it because we we have a partially ripped up file so the modeling of what is a hidden State versus an observable is really just a question of like what do you have data for or want it and then what do you want to include as a causal or latent factor in your model but you don't necessarily have data for as an observable but you may even choose to have hidden States for things that you could get data for

so I think that's the kind of deflationary bent in this textbook and and maybe in many people's active inference Journeys when the model itself is built as they encourage people to do from the first chapter to the last then you see the specifics of how it plays out in that generative model and the contributions to the field consist of specific generative models made and general claims with the active inference ontology

the thing

um that it struck me is to in order to act in the world to remove myself from the temperature that's going to burn my skin I don't actually have to know anything about the generative process I can make assumptions about the generative process and then act in the world to try to alter that but my my confusion lies with how we can then

within model construction

dictate what the process what the

generative process is and I can

understand why you have to do that if

you've got to create a model but it kind

of just doesn't speak to

the

to the the ideas laid out in chapter one

in this book you know that's that's

another angle is uh when you approach a

specific question or a line of inquiry

you're not trying to make the the total

um twin through a scientific model

you could learn something about the knee

without having the full

simulation whatever that would even mean

of a knee

so

I

in the specific modeling but maybe

that's coming from an experimentalist's

perspective so I'm not sure Alexi

yeah I I had I had maybe in a reaction

but I wonder if maybe we need to maybe

time to maybe let this idea sink is it

appropriate for me to ask another

question on a slightly different topic

or would that be too abruptly yeah

definitely that sounds totally good if

you are if it's here or just ask it so

it's about chapter one and the uh I'm

reacting to this text action is

quintessentially goal directed and

purposive

okay so

um I understand that you know there's a

goal to stay alive no questions right

but you know if we move to human

subjects and you know uh uh uh please
forgive me sort of Clinical Psychology
Psychotherapy and there's a ton of
action that humans do that at least
doesn't appear as goal directed or
pervasive in fact if every action humans
did was goal directed and purposeful if we
wouldn't need Psychotherapy at all
like we have I don't know

um
maladaptive behaviors we have obsessions
compulsions ticks and whatnot so uh I'm
not exactly sure if we stand with that
postulate that every action is going to
essentially goal directed and pervasive
would that model not oversimplify things
for human subjects

that's a great question a short thought
on it and then anyone can raise their
hand is

even an isocade can be purposeful
purposive because you can explain it in
terms of using this framework so it
doesn't necessarily mean that in some
integrated way it's beneficial for the
person's life or for any given social
recognition or bodily Health necessarily
because you can do modeling of
pathologies too but the purposiveness is
just local teleology local directedness
moving the elbow from one to to do a
slot machine again and again is
purposive you can argue whether it is
with the person's long-term professional
interest but the action is still
purposive

yeah well my my issue with the
teleological explanation is that it is

often uh very speculative when people
for example tell me that my state of
depression has a purpose to inform me
that I have an unmet need which is
mainstream you know models and Clinical
Psychology sometimes like how do you
know that depression doesn't have five
different purposes and models we pretty
much you know if you if you say in first
speculate that that's the the purpose of
depression so the teleology is is a
little kind of weird for me maybe it is
mainstream and active inference
framework but it's a little weird yeah
but maybe I'll think about it more a
related question is this when we say the
goal directed character of action the
question is whose goal again going back
to human subjects there's conscious
there's unconscious processes and
sometimes people do something and
they're like I did this thing I
rationally know I shouldn't have done it
but I did it anyway
which suggests that they have multiple
components of their agency and not just
you know a unitary John who you know
decided all the goals and purposes so
the question is whose goals what what
exactly do we assume unitarity do you
assume that the agent is a singular
thing
yeah great question Kate
well my thoughts on this are that
um I would say that in my understanding
at least
um
active inference views the Mind as a

hierarchical structure but definitely
hierarchical buildup or many different
um
sub layers of the brain and ultimately
the goal of therapy like in some random
therapist but in some way it could be
the integration of all these different
sounds but those in by themselves the
lower lower levels or lower parts of the
hierarchical components of the hierarchy
within their own domain they are
purposing action the action within those
domains they don't they don't solve the
host necessarily as such but within
their own logic internal logic even if
you could put it in the math it would
fit the free energy principle and the
active influence so the goal there is to
figure out how to bring the hierarchy
together in a more purposive or healthy
way that's that that's the thought that
your comment triggered Alexi that's all
I have to say about that I I'm very much
for this thought and it's beautiful
except you know bear with me maybe I'm a
devil's advocate but what I see in
clinical work and what is mainstream and
psychodynamic Psychotherapy is mental
conflicts such as I want this
but I cannot have it or I I should be
doing this but I want to do something
else or I Want A and B and they're
mutually exclusive and in that regard
there is no hierarchy there is an
unresolved conflict mind and a conflict
is very mainstream kind of 100 year old
clinical tradition and you know
ultimately you don't resolve those

things you know that that mind goes from
one conflict to another to the third one
we have multiple needs like the Famous
Madonna syndrome is an example of
that and how people find a compromise
from that is a sort of personality so I
I just do think that some of these
assumptions and postulates we make early
on that there's this you know either
it's resolved through the hierarchy or
there's a singular agent whose goals and
purposes we follow are you know okay for
for a bacteria but maybe not for human
mind

well but um card Horace Harris
um his studies on psychedelics
and the um his theory of how the
psychedelics would kind of reset the
priors reset the Precision of the priors
the villains and in many cases it
oftentimes helps with some of the
symptoms which would argue for that view
so you may not be able to do it verbally
through a verbal interaction or through
language but in some other you know
parts of it may be open to more chemical
or you know

neural transmitter
kind of related reset as opposed to a
verbal reset

but anyway

just just to kind of

um give it one thought and then ali
um goal is not really a required term
just my personal view in the active
inference ontology we don't talk about
kicking a ball through a goal we don't
talk about a milestone based goal it's

it doesn't come up when we're actually using the active infantology it's not a parameter in the model but cybernetics has long been antilios and all these debates so I think actually what's really exciting and I'll say here for now is it actually points towards exciting different ways of understanding purpose of behavior and where goals and rewards come into play by thinking about likely paths being selected rather than rewarding paths and how that generalizes across a system that stays in a wooden Groove and how that also May relate to these higher order sophisticated cognitive functions like narrative but that is not something that we're going to see in figure 4.3 that's something where once you build that Tower of Babel then you'll know but hypothetically what can you say without the generative model of what people are talking about because we're not talking about just the base four nodes modeling the whole brain like that if you were going to make a model that had multiple debating personalities then it simply would have that feature

Ollie

well actually I think uh the term action here

is used

um instead of in I mean to refer to intentional action

so I I don't think it's action in its broadest sense but even if uh even if we restrict ourselves to only intentional

actions there are some theories both philosophical and in cognitive science uh that tries to somehow frame this notion of action in terms of an alternative conception of causality so one notable work in that regard is the work of Alicia

in her work in her book Dynamics and actions so in this book I mean the basic premise of this book is exactly uh the question what is the difference between a wink and a blink right so uh

I mean in order to come in order to come up with a viable theory of action that that can include all of these uh intentional and non-intentional actions uh she proposes a theory of causation based not on uh traditional linear cause and effects but rather

[Music]

looking at the whole complex adaptive system as um as as possessing some constraints and looking at those constraints as uh as as somehow giving rise to causation so you know the words the what we mean by causation uh is actually undertaken by those constraints of those uh complex adaptive systems sorry so these dynamical constraints uh can actually account for both bottom-up and top down causal relations so in this sense it can include both intentional action and non-intentional action and if we take goal-directed

action as uh kinds of action that tries
to satisfy those dynamical constraints
then this

um I mean claim in the book can make
sense in this

complex dynamical system framework
great thank you Lexi

I just had a couple reactions uh Daniel
to what you said and thank you for your
comment about gold being necessary or
not please forgive me if my metaphors
are very simplistic so the car sitting
in the parking lot idling burning gas
is increasing entropy it is uh not
moving anywhere right a car going to
where I needed to go is going toward a
goal and perhaps it is wasting some
energy and as well so if we remove the
goal out then you know how will we
actually look at the free energy versus
you know the energy used toward a
meaningful purpose you know so I think
implicitly perhaps a goal like the goal
is to survive for an organism and to
maintain survival evolutionarily survive
until reproductive agent reproduce you
know some kind of goals like that exist
perhaps you know if you take all of them
out then then it's all entropy right and
also had a maybe reaction to Kate that
you know I am very very respectful of
Cartwright Harris's work and
psychedelics with that I just I'm highly
skeptical of the idea of reset or
anything brief because you know
attachment style in a human being takes
from 8 to 66 to uh 18 months you know
nine women pregnant for a month will not

produce a baby you know personality
takes time to form and all of this sort
of quick you know uh resets whether it's
Ayahuasca or ketamine or even
electroconvulsive therapy I just don't
buy that you know I think that that it's
minded and conflict remains in a
conflict it just how it solves the
conflict is is the question it may be
solving it more adaptively more you know
in a more nuanced way but I don't think
that anything quick like a magic wand of
psychedelics will will resolve that in
any way so

yeah one paper by Adam saffron is the
Albus model so while that relaxation of
beliefs was more visible
and led to a kind of
um relaxation based emphasis on what
psychedelics might do which may be half
the picture

um Adam just kind of generalized it to
altering like well depending on which
generative model you make or how you
think about what the brain is or
whatever

um different

um pharmaceutical or non-pharmaceutical
experiences are going to not just purely
relax a distribution because the
distributions are part of the map

so I think

um like

it it's not a property the physical
molecule to do so except again
interpreted in this more abstract causal
space

um

to your question Alexi

um

you know you talk about a car in the parking lot it's great it's just disordering material and it's not going anywhere it's not goal oriented but its purpose could be described hyper locally as just being in one place like a rock's purpose just using again this broader concept of purpose which doesn't come up in the acting phonology we don't need the goal but the Rock's goal is just to be right there the Leaf's goal is to go Downstream so by having a deflationary optional concept of intentional Behavior

including um like Carl first

um wood lice example

where there was insects running around on the pavement and they went faster where it was light and slower it was dark

so they accumulated where it was dark so including a whole Continuum of intentionality

based upon return to an attracting set is it just turns out to be enormously General

now how to connect that to the the physical entropy lost by organisms open question

and then how to use some other um terms from thermodynamics like what would informational work be versus informational Heat

well Chris Fields is certainly approaching those types of questions but um

I hope that addresses a few pieces of it
something doing the most likely thing
is exactly the Criterion that you would
use first for statistics
and so in that way it's why we can use
statistics so
um adeptly in this modeling
because statistics isn't about simply
maximizing reward but like all
statistics is about finding
um this path of least action in an
informational space Ali
sorry it was my hand risen from before
sorry oh no worries
anyone who hasn't uh brought something
up just definitely has gone all over
which is totally
cool but if somebody wants to go to a
specific question of chapter one or a
quote we could do that
um I did want to chime in on the
the previous topic about
um
action is quintessentially goal directed
and Professor uh
so I mean you know we have some ways to
get further into the textbook but I mean
we're going to be going over things like
policy selection kind of like I'm
assuming active inference involves not
just settling upon a single unchanging
policy right continuing with that that's
what an agent operates with rather it
would be you know potential entire
series of competing policies that may
very well be actively updated based upon
what the agent has maybe learned in the
past new incoming observations so I'm

just kind of reflecting on everything
that's been said and
um because whenever I read this I I
would it feels a little bit easier for
me to take uh the phrase goal directed
with a bit of a grain of salt and think
like oh there could be multiple
potentially even conflicting policies in
play that themselves are constantly
potentially changing or shifting a
variety of ways so just yeah it's just a
kind of final comment they supposed on
that topic I was thinking about yeah
that's awesome it's it's so true policy
is evaluated every time steps so this is
exactly how we integrate
action into the loop whether like ooda
Loop or just behavioral uh loop so
that's a great point

Terry

and just uh the way I interpreted that
was that the goal isn't uh that the goal
is fundamentally to
um answer questions of the universe and
uh therefore

uh

get evidence that then can be fed into
the variational free energy model
uh to either update or prioritize or act
in the world to

um

to reduce the variational free energy so
the goal isn't

a goal to achieve an end the goal the
end is to

answer the question about

uh

minimizing variational free energy

and then our behavior in the world that
we read as
policy LED goal directed is actually
more a function of that primary
objective of the organism which is to
minimize variational free energy
great points
again I'd highlight that we can Model A
baseball on a parabola as doing the most
likely thing we could identify a
radioactive isotope that's
probabilistically emitting things over a
given time window as doing the most
expected thing
so a lot can get brought into statistics
which does breed this whole fascinating
area of like map territory type
distinctions
so suffice to say action is going to be
totally integrated into this model
enabling a lot of interpretations
as the recent literature Witnesses Lexi
thank you so much Daniel I was just yeah
thinking about everyone everybody said
but I think your word intentionality
connected it from me and maybe the issue
that I have please forgive me is lexical
because so in my world you know
intentionality is required the the
sentence you said that Rock's goal is to
stay in place is an oxymoron and and
sort of psychology because you can only
have a goal if you have a mind
so in that regard bacteria that moves
toward the like you know gradient of the
food and withdraws from Pain does not
have a goal and a purpose because it
ain't got a mind so that that maybe is a

different use of language and if they use the word gold here in a sense that the Rock's goal is to stay in place it's an attribution I would perhaps it's an unfortunate use of a term but maybe we can say It's You Know The Rock's Dynamics started to stay in place but it doesn't have any goals you know there's no mind there there's no agent there's no agent capable of choosing a versus B in Iraq yeah

I I I totally agree like it is sort of a in a narrow and a broad sense sometimes there's going to be like gray area assignment just like um well this enzyme wants to bind to this one I've heard it a thousand times so it's like but it's not intended as a reflexive

you know statement about the protein being a strange Loop maybe it's below some level of critical ability but we can model this all in an integrated fashion so we can still model the game theory of something that we don't even think plays it implicit

representational game that's why there's this whole representationalist debate and then we can do all these kinds of modeling links really composably which is kind of the exciting piece which is also why the the kernel

ultimately what we're working towards like figure 4.3 these generative models the kernels vary

it's it's not applicable to any given

situation except a purely purely
pedagogical
and it's extremely sparse on
commitments
because it's basically of the same type
as every other statistical model in
Bayesian statistics
so it's not like an esoteric type of
model
but it opens many questions that I think
again we see every day which is really
exciting and part of the fun and and why
many people can contribute in as well
but there's just so many interesting
angles to it and I think also sanjeev's
textbook
that um is in progress with several of
you um here who who may be um in that
group like
that there's a whole statistics degree
slash
undergraduate or Beyond leading up to
a lot of things that really help set the
stage
and and his his work I I think will be
like for people who want to see that
level of depth
and then meanwhile understanding what
other
um fun paths and ways are there in this
whole
work because certainly it's always every
sentence you could be like
what is happening and uh you know if
this word meant that and if this person
was defining it this way
so certainly finding a path on through
and like what's the claim

when a lot of it is internally

referential

um

however that's part of its coherence and

consistency

and those the but sometimes the evidence

for the consistency is a little bit hard

to

convey because these are relatively

recent

understandings

hopefully that's just like a little

context

foreign

anyone else have a thought or a question

on

chapter one Terry or than anyone else

and

it's it's relevant to this thing about

that a rock has no mind but a rock guy

is you know maybe over longer time

scales to sustain itself as a rock it

has to uh

minimize its free energy in terms of its

uh interactions with the external State

and whenever the uh variational free

energy within that system Alters it will

never be oxidize or you know get broken

up or if they think you know it it can't

act in the world to maintain its

variational free energy at a minimum but

it still has to maintain that

variational free energy at a minimum and

when it can't it changes to something

else so you know the notion that because

that that that

um

you know all objects and in the world

sustain themselves and they're not in in
the state that they are currently in by
minimizing variational free energy
whether you're a tree a human a rock or
a you know a Galaxy probably

um uh I I what's your reaction to that
because I I know that we we you know
they reference

like trees aren't can't do things but
trees do communicate with each other you
know as we and and I suppose I was
obstruct by this concept of pan psychism
that exists within you know the sort of
the the the thinkers in in in in in in
the nature of Consciousness I'm just
interested

yeah it's it's something that's
stimulated by what Alexi said and I'm
interested in what you have to say about
that

yeah

it's very it's interesting I mean it's
certainly interesting areas you brought
up and the way that there could be a
kind of deflationary account
where just it's pan cognitivist
or just pan computationalists but then
people debate what's a computer and they
say well it's pancognitivist but then
they debate what is cognition is the
baseball Computing and it's the water
Computing where it expects to be
downhill so those are all the debates
but

most of us

um

see a world with dense but sparse
connectivities

so

it sometimes it's like a lot of problem
admiration which can be a good thing
but um pointing to communication
Pathways amongst living things
is only surprising in light of extremely
recent specific scientific observations
so but I'm I'm not sure I mean that's
there's so many other areas you brought
up in the Consciousness and I mean these
are all the huge
questions Ali

uh yeah again uh one of the other common
misconceptions about fep is that some
people think fep as providing a kind of
ontological conditions for existence or
at least persistence through time
but actually what fep claims is that if
something exists it must have properties
that look as if they follow the path of
least action so it's the other way
around fpp doesn't provide any
conditions for the Existence or
persistent through time at all but it
rather I mean if we take the persistence
or the existence as the basic premise of
our argument then we can
and follow that by consider by
describing the behavior of uh of the
phenomena or or the system by describing
uh the
um I mean trajectories of their
variational density uh in terms of uh
least action principle so uh I think
this is one of the things that people uh
some sometimes
criticize fep as something that I mean
of course it's

claimed to be theory of every thing but
it's not something as some ontological
or metaphysical
theory for anything it just tries to
describe I mean beginning from the
ontological existence it tries to
describe the ensuing characteristic six
of the systems that persist through time
and in particular
highly complex
high-dimensional self-organizing systems
amazing points Ali thank you
the condition of modeling something
is to have these properties every
measurement that's that's if it's static
it's it's uh with its movement is within
the measurement the classical limit
if it's so Dynamic it's moving faster
than the time scale measurement you just
get basically uninformative signal so we
always kind of tuned in Bucky Fuller
zone of modeling Proxima modeling which
we can extend with microscopes uh
telescopes
um that's one way to look at this that's
a kind of like scientific realism
approach
but then for things that you can't or
didn't model you have hidden States
so ultimately you have it covered
um but then this is really the the
inversion of the um ontological theories
of everything that this is how it really
is this is how it is out there
um this is how the metaphysics are that
you know project our world a certain way
that is like why ontology is the
philosophy of what is

however if we precondition upon
existence someone will look at an and
wanted to go off the trail there it's
like
but you have the whole evolutionary
history that sets up its decision making
so maybe the purposiveness of that you
might have a super deflationary response
um but again whatever level of
sophistication the thing is from
essentially inert
or totally inert to repetitive motion to
sophisticated motion
that can be understood by an observer as
a path of least action
as they do statistics
so in that sense it licenses little more
metaphysics than a linear regression
as an analytic method
then I think again there's so many
interesting angles why it has been
um discussed in all of these more
philosophical contexts
um
because reductionism isn't the whole
picture
so the theory in itself or any given
theory in itself might not take on all
these accounts
and there may just be tangled webs of
perspectives and and difference
linguaging and uh different applications
around some of these topics
but it is fun and fast when it's small
and the few people's perspectives are
essentially knowable
and there are
um always new information coming in

literally through these funny modern
channels and hearing what Maxwell and
Carl at all think Andrew
yeah yeah I just uh just my two cents
there is that um
you know I find the pan cognitive stuff
um rather interesting than that said I
personally came to active inference
because I'm interested in its
application to uh living systems and
especially human beings and so the brain
uh from a neuroscience perspective but
also how it might be applied
and working with psychopathologies
treatments and so on and so
um I think so far I'm just viewing it in
more pragmatic sense of like how can we
use this in the same way that I also
work as a data scientist and so I don't
view everything as a gradient boosting
machine but I will attempt to apply a
model to a situation to see if I can
get anything useful out of it right so
that's that's kind of how I've taken
this so far I mean I've read some really
fascinating papers you know one by a
psychiatrist who applied active
inference to uh post-traumatic stress
disorder described um hyper precise
priors in the upper cortical regions is
like you know someone who experiences
reliving the
um excuse me that can be triggered by a
certain phenomena that kind of spark or
activate those those previous priors
that gets superimposed on the incoming
sensory observations even though it's
inaccurate

um so that's I mean
I'm personally a little less concerned
with the Precision of ontological claims
and so on but um but but kind of
how this can help us to reinterpret
different kinds of phenomena and from
there maybe use it like for good for
some kind of useful purpose and sounds
really it's really fascinating you know
so it's

awesome thank you for sharing that
yeah

it is very interesting
anyone else want to just give a comment
in the last
minutes

or otherwise we will be

um

going into chapter two

chapter two is gonna be uh

another two weeks

we can look at the figure and look at
the high road and low road again just
kind of look ahead

low road

and the high road

chapter 2 and chapter three that will be
the coming month

also some people

um

have asked about the

math learning group

so

I I made a little comment thread if you
just comment on this thread then I think
people who want to do a certain kind of
work can maybe find each other this way
and it can be in the Discord or a gypsy

link or however people prefer

um

and uh

yeah continue to to check out and

comment or or improve if you want any of

the AI

math things as we head into chapter two

um

explore which of these equations if you

can evaluate which one of these AI

explanations or make comments on them

some are useful some are not so

you know

but I hope there's a lot of fun things

we can do in the coming

weeks chapter 2 is definitely going to

get into it because it just starts the

real non-introduction part of the book

um

but yeah does anyone have any other

thoughts or questions or anyone who's

read chapter two before

Wanna Give a teaser

or or

Ali go for it

well yeah actually chapter two is where

the real fun begins I mean after

the introductory chapter and now we're

uh into the real set of things by

considering the two of the most Central

equations of active inference namely

equations 2.5 and 2.6 one for

variational free energy the other four

expected free energy which will come to

time and time again throughout the whole

book

so it's probably one of the most

fundamentally important chapters

uh

great comments anyone else want to add

anything

okay

all right thank you all see you around

bye